

Technology Applications in Teaching at the Right Level Programs

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Abstract—Around the world, inequality in education is creating difficulties for students as they learn how to read and write. Low resource and rural communities in East Africa, and Kenya specifically, are facing issues with students not achieving the national standards for literacy competence by the time they finish primary school, leaving them unequipped to progress in their education. Many groups have been combating this issue, with one particular methodology making a large impact in early childhood literacy and numeracy, known as Teaching at the Right Level (TaRL). By learning in small groups with interactive instruction, students' weak points are addressed before moving on to more difficult concepts. However, this methodology is traditionally carried out using only paper, increasing the time needed to collect and organize student results and data. To combat the inefficiencies of the traditional method, we have prototyped a digital platform that is able to accurately assess student literacy levels, seamlessly input and organize data as well as provide a library of learning activities to instructors. This platform will introduce students to using technology as an aide in learning as well as increase the efficiency of this learning process, hopefully leading to more students achieving higher learning outcomes.

Keywords—literacy, education, technology for literacy, education technology, Teaching at the Right Level

I. BACKGROUND

Across the globe, millions of students face educational inequalities that cripple their ability to build foundational skills and create barriers to taking advantage of career opportunities. Specifically in Kenya, primary school students are unable to complete coursework intended for students even younger than them; this disparity is especially prevalent in rural areas. To reverse this trend, non-governmental organizations have adapted a learning methodology called Teaching at the Right Level (TaRL) to help students who have fallen behind in their learning quickly progress in their core foundational skills. While TaRL is highly effective, there are inefficiencies which hamper higher learning outcomes for students. There is a large opportunity space for technology to make instructors more efficient and increase instructor-student interaction in TaRL programs. In this space, Nyansapo, a Kenyan edtech company that is an outgrowth of a collaboration between Penn State's Humanitarian Engineering and Social Entrepreneurship program and LakeHub Kisumu's accelerator programs, is developing a platform that automates the rote tasks of TaRL to optimize an instructor's process to increase learning outcomes for students.

A. Educational Inequalities and Barriers to High Learning Outcomes

The United Nations Educational, Scientific and Cultural Organization (UNESCO) estimates that 250 million children cannot read, write, or count well, and 200 million young people leave school without the skills they need to thrive [1]. In lower and middle-income countries worldwide, students in the same grade level tend to be at a wide variety of learning levels. Towards the end of primary school, many students are far behind the expected level set by the curriculum [2]. Without these basic foundational skills, students are unable to effectively progress through their education and are afforded less opportunities later in life to attain a career that provides a living wage.

In East Africa, over 18 million children do not complete their primary school education. [3]. Specifically in Kenya, a rural classroom averages about 40 students under the direction and supervision of only one teacher [4]. This can be linked back to the introduction of the Free Primary Education Act of 2003 which resulted in Kenyan public schools seeing an increase of 1.3 million students from 2008 to 2012 [5]. This increase of students was not met sufficiently with an increase in teachers. In rural Kenyan schools, libraries are rare, and it is common to find students sitting at overcrowded desks and using few textbooks, notebooks, or writing utensils. Due to this lack of resources, seven out of ten Class 3 students cannot complete Class 2 work [4].

B. Teaching at the Right Level (TaRL) Process

To catch students up in their learning, non-governmental organizations in Kenya have adapted the TaRL methodology developed by Pratham, a non-governmental organization based in India, to supplement students' learning outside of their usual classroom instruction time [6],[7]. The TaRL process turns disadvantageous aspects of schooling into assets - utilizing high student-to-teacher ratios, sidestepping a lack of classroom materials, and building a program around aiding students at their current instructional level.

The process begins with individual students' assessments that are used to gauge the skill level of each student. The instructor meets with each student individually and follows a standardized assessment protocol in a friendly environment. Being a standard part of the TaRL process, assessments provide

an opportunity for both instructor and student to engage with each other and personalize the learning experience.

Next, instructors use the assessment results to group each student into a learning level group of their peers. The skill levels are pre-defined based on the subject area and are irrespective of age or grade level. Large class sizes are taken advantage of by grouping students, so students can learn from their peers and socially interact with each other - an important distinction from traditional classroom learning.

Finally, with students in groups, instructors develop activities that are targeted to each general skill area. For instance, in literacy, there are six learning levels: first time learner, letter, word, paragraph, story, and above. With students in groups, the stress on the teacher to manage different skill sets simultaneously is alleviated; each group follows a different instructional plan that is fitted to their learning needs.

The process of assessing, grouping, and using activities is repeatable until all students make learning gains up the 'skill tree.' By operating in a cyclical fashion, TaRL can be used to advance students' present learning needs and build on them - a form of instructional scaffolding. However, there are inherent inefficiencies in TaRL that can make it ineffective in some settings.

C. Inefficiencies in the TaRL Process

Based upon personal experience with TaRL, several inefficiencies were noted. At the beginning of the process, instructors must spend upwards of one to two program days assessing students. Instructors find this process strenuous and time-consuming. Assessments absorb time that could be better allocated towards making learning gains further along in the program.

The student learning groups are static and immutable until the next assessment round. Since students may be at different sublevels within one skill group, there is no opportunity to cater to needs within a group. A student that has mastered a skill group must wait until the next assessment round to advance.

Finally, the developed activities that instructors present to each group are targeting the general skills of that group. Instructors are not able to target specific sub-skills that students in a particular group may be lacking.

D. Potential of Technology Use in TaRL

A 2019 study about integrating TaRL and technology has found that using technology as a complementary tool for an instructor to automate rote tasks and process student data for instructional insights shows immense promise [2]. Such a tool would shift more of the instructors' time and energy onto their students and increase instructor productivity.

In TaRL's present state, there are few tools for an instructor to identify, address, and monitor each student's zone of proximal development in real time. Due to the need to assess all students one-on-one and record all student data manually, instructors lose valuable teaching time. These inefficiencies affect instructors' abilities to provide valuable learning experiences as well as the students' abilities to get the most out of the external help being provided. These problems are inherent to any TaRL implementation.

E. The Nyansapo Platform

To optimize an instructor's process in TaRL programs, Nyansapo has developed an app with three main components - digital assessments, dynamic grouping, and a library of activities. The app was developed using Kotlin in Android Studio[®] with backend services provided by Google Firebase[®] and Microsoft Azure[®]. User data was also stored and encrypted on Google Firebase[®] [8].

As shown in Fig. 1, instructors administered assessments digitally through the app by creating a student's profile, selecting an assessment, and presenting the mobile device to a student. Before the digital assessment began, students selected one culturally appropriate animal avatar (e.g. lion, water buffalo, elephant, rhino, or cheetah) that read instructions aloud in the students' native language of Swahili and explained the instructions for each stage of the assessment. In addition, instructors were available to assist students if any questions arose.

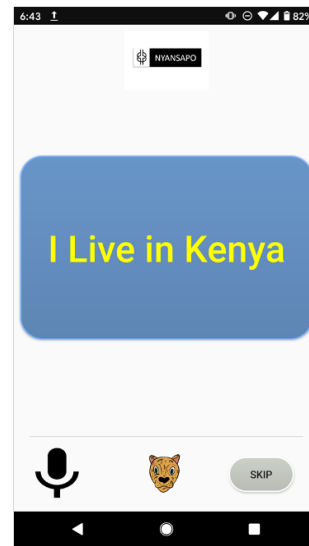


Fig. 1. Student view of the app's assessment feature

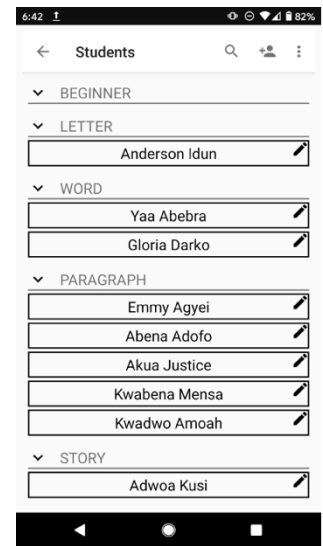


Fig. 2. List of students from the instructor view

After assessing all students digitally, instructors can access the learning level grouping assignments as seen in Fig. 2.

This instantaneous process stands in sharp contrast to the arduous process of individually administering assessments, recording the student's learning level, compiling the data, and handcrafting each group.

In addition to providing grouping assignments, the app presents insights on an individual student's progress as well as class-based statistics such as commonly missed words in a literacy curriculum.

Also, the app provides an interface with which instructors can formulate Session Plans using preloaded activities as well as giving instructors the ability to create their own, allowing for use in different contexts and groups. Fig. 3 presents a selection of literacy activities for the paragraph and story level. Instructors can quickly access a list of activities and assemble a session plan from scratch.

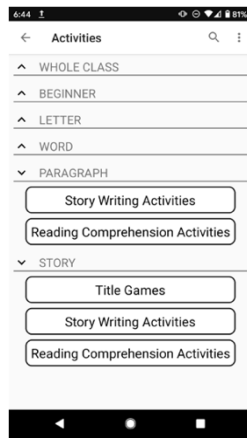


Fig. 3. Activities page showing suggested activities for each literacy level

For TaRL programs, the Nyansapo platform claims to offer a technology that aids the learning process by digitizing assessments, streamlining routine tasks, and automatically analyzing data. Complementing programs with the app can increase instructor-student interactions that provide students with more opportunities for better learning outcomes.

II. THE PROBLEM

The traditional implementation of TaRL programs heavily relies on instructors to complete repetitive and strenuous tasks, manually assessing each student individually in addition to recording and compiling data on the entire class. These tasks take away time from the activities section of TaRL, where students make real learning gains, and ultimately dilute the value that students can receive. There is a need and an opportunity for these tasks to be automated by a technology tool such as the Nyansapo platform. With a digital tool that optimizes the instructor's process and allows them to focus more on their students, TaRL programs can be made more efficient and produce better learning outcomes. The balance of this paper looks at whether the Nyansapo app is effective in that role.

III. RESEARCH METHODS

Kenya was selected as the testing location partly due to the high technological literacy as compared to other similar countries. Kenya leads the continent of Africa with 91% smartphone penetration, as compared to the continental average of 80% [9]. There is also a push from the Kenyan government to provide technology for children in primary schools, with over 1 million tablets being distributed in 2020 [10]. Kenya provides the widest market for a technological solution to this issue. To test the effectiveness of this technological tool, three sequential literacy boot camps were run with primary school students in Kenya. The goal of this testing was to compare two methods of the TaRL process - traditional and digital. The traditional process uses paper to assess students and heavily relies on instructors to organize and collect student progress data manually. The technological solution assesses students digitally using the Nyansapo app on a mobile device and automatically collects and tracks student data.

A. Custom Speech-to-Text Model

For the literacy assessment, the Nyansapo platform provides a custom speech-to-text model made specifically for Kenyan primary school students. The model was created by visiting primary school aged boys and girls from six counties in Kenya and obtaining recordings of their responses to sections of a sample literacy assessment with permission. Over 3,000 recordings of letters, words, and paragraphs were collected, individually annotated, and then inputted into Microsoft Azure's Custom Speech Portal[®] to create a cloud-based model with an accuracy above ninety-five percent [11]. When a student took an assessment on the phone, the smartphone prompted the student to read aloud a letter, word, or sentence. The phone's microphone recorded and transmitted the audio recording to an Azure server for transcription. Then, the transcribed audio string was sent back to the phone. By comparing the transcribed audio string to the string of the original reading selection, the app determined if the student made any mistakes.

B. Boot Camps

Three different boot camps were run over the course of 2019-2021. The first bootcamp, held in Voo, Kitui, Kenya during May 2019 excluded all forms of digital tools for the purpose of understanding the development, operation, and success rates of a traditional TaRL program. The second bootcamp, also held in Voo, in August 2020 utilized the Nyansapo Android[®] app platform with a digital literacy assessment in place of the traditional paper assessment. In April 2021, a third boot camp was conducted in Korando (near Kisumu, Kenya) where students were sampled into two different groups - an experimental and a control group. The experimental group was assessed only using the digital assessment on Nyansapo's platform while the control group was assessed using the traditional, paper assessments. The learning level data, gathered from the April 2021 boot camp is the focus of this paper.

C. Boot Camp Preparation

The preparation process for each boot camp consisted of five core parts. First, instructors were recruited locally (often schoolteachers or college-aged recruits) to participate in the boot camp.

Second, all instructors received training in the TaRL methodology covering the fundamentals of curriculum (assessment protocol, grouping, and activity development) and how to create day-to-day session plans. In addition, instructors were provided Android[®] smartphones with the Nyansapo platform pre-installed to become comfortable with the user interface, learn how to administer assessments, and analyze results.

Third, instructors began the student enrollment process. Each student was assessed using the literacy or numeracy assessment (depending on the focus of the boot camp). If the student's learning level demonstrated need, they were designated as a potential enrollee. Often, instructors who are primary school teachers during the day assessed their students during class breaks such as game time. Otherwise, recruited instructors traveled to students' homes in the area, discussed the

boot camp with parents or guardians, and assessed students at home.

Fourth, a location for the boot camp was secured. Most often, the location was a schoolhouse or a communal gathering place that was local and familiar to the instructors and students.

Lastly, all the materials needed for the boot camp such as arts and crafts supplies, textbooks, and writing utensils were transported to the location in preparation. In light of the COVID-19 pandemic, the boot camps in August 2020 and April 2021 primarily conducted activities outdoors with masks and hand sanitizer provided for all students and instructors as shown in Fig. 4.



Fig. 4. Two students in the same group working on a literacy activity together

D. Day-to-Day Boot Camp Protocol

Each boot camp session was conducted either during school hours (breaks in the afternoon) or after school hours. These sessions lasted for approximately one hour and were conducted for ten days. Each session began with recording student attendance.

Because each boot camp focused on literacy, each camp had a pre-assessment section and a post-assessment section that used the Uwezo literacy protocol [12]. In this protocol, each level of the assessment has criteria that students must meet to pass.

Each student begins the assessment at the paragraph level. If students read the paragraph and mispronounced less than four words, they are moved on to the story level. Otherwise, if the student mispronounced four or more words, they are guided to the word level.

If a student did not pass the paragraph section, then the first section that is passed becomes their learning level (e.g., letter, word) or they are placed at the beginner level. If a student passed the paragraph section, they arrive at the story level. Then, they must successfully read a story and answer two comprehension questions to pass the story level. If the student mispronounced five or more words while reading the story, the student is placed at the paragraph level. On the other hand, if the student successfully read the story with five or fewer errors

and failed to answer the comprehension questions correctly, they are placed at the story level. Otherwise, they are assessed to be at the Above level, which is equivalent to a fully comprehensive English reader at a Grade 2 level. A flowchart of how students can progress through the literacy assessment is seen in Fig. 5.

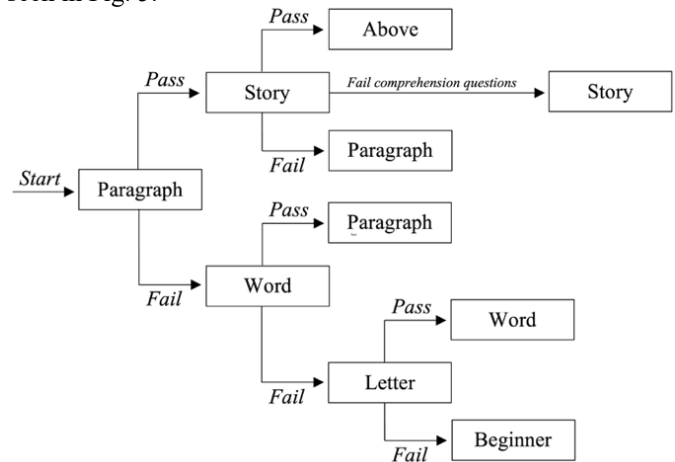


Fig. 5. Flowchart of literacy assessment progression

Students in the experimental group were assessed using the Nyansapo digital assessment. Students in the control group are assessed using the traditional paper assessment. The student's learning level and group (which, in most cases, are used for the entire duration of the boot camp) are determined by the level achieved from the pre-assessment.

After the pre-assessment, students are grouped according to their learning level. Students' placement in the experimental or control group for assessments had no influence on their placement in a learning group. In groups, students are engaged in educational activities aimed at their literacy level. An example activity session plan is seen in Fig. 6.

Day	Whole class	Letter	Word	Paragraph	Story
Day 1	-Sing Hello Song (10 minutes) -ABCD song in the middle -Goodbye song	Alphabet cards (10 minutes) Find partner capital and small letter - (10 minutes) Letter Treasure Hunt (10 minutes)	Word formation (15 minutes) Cards with letters to form words (15 minutes)	Treasure hunt (10 minutes) Mindmap of the words introduced (20 minutes)	Complete the sentence (10 minutes) show me (20 minutes)
Day 2	-Sing Hello Song -Read aloud -Goodbye song	Find partner capital and small letter	Replace a letter	Pictionary	Show me
Day 3	-Sing Hello Song -Read aloud -Goodbye song	Who comes next	Find the word	Riddles	Sentence puzzle

Lesson plan by Esther

Fig. 6. Example of an activity session plan for a five-day boot camp with two days of assessments and three days of activities

The final session of the boot camp is the post-assessment section where students are reassessed to determine whether their learning level changed. The type of assessment given to each student (i.e., digital or traditional) was determined according to their placement in experimental or the control group from the start of the boot camp.

IV. RESULTS

The April 2021 literacy boot camp was conducted following the TaRL methodology and included 47 students. 22 students were randomly sampled into the control group where they were solely assessed on the traditional paper method. 25 students were sampled into the experimental group where they were assessed using the digital assessment platform.

Fig. 7 and 8 display the distribution of students' learning levels in the control group and experimental group, respectively.

Control Group Distribution of Learning Levels

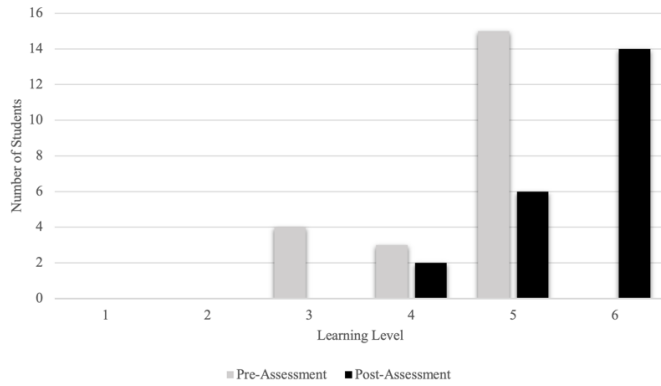


Fig. 7. Change in control group learning levels before and after the boot

Experimental Group Distribution of Learning Levels

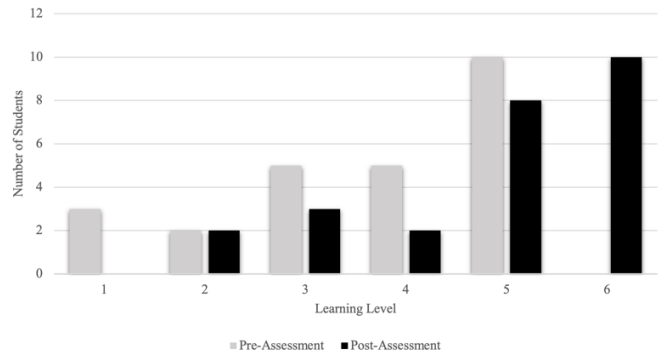


Fig. 8. Change in experimental group learning levels before and after the boot camp

Tables 1 and 2 show the numerical distribution of students' learning levels in the control and experimental groups, respectively. Tables 3 and 4 show statistics on the control and experimental groups, respectively.

TABLE I. CONTROL GROUP LEARNING LEVELS

	Literacy Levels					
	<i>Beginner (1)</i>	<i>Letter (2)</i>	<i>Word (3)</i>	<i>Paragraph (4)</i>	<i>Story (5)</i>	<i>Above (6)</i>
Pre-Assessment	0	0	4	3	15	0
Post-Assessment	0	0	0	2	6	14

TABLE II. EXPERIMENTAL GROUP LEARNING LEVELS

	Literacy Levels					
	<i>Beginner (1)</i>	<i>Letter (2)</i>	<i>Word (3)</i>	<i>Paragraph (4)</i>	<i>Story (5)</i>	<i>Above (6)</i>
Pre-Assessment	3	2	5	5	10	0
Post-Assessment	0	2	3	2	8	10

TABLE III. CONTROL GROUP MEANS AND STANDARD DEVIATIONS OF PRE- AND POST-ASSESSMENTS

	Pre-Assessment		Post-Assessment	
	<i>Mean</i>	<i>Standard Deviation</i>	<i>Mean</i>	<i>Standard Deviation</i>
Students' Literacy Level	4.5	0.78	5.55	0.67

TABLE IV. EXPERIMENTAL GROUP MEANS AND STANDARD DEVIATIONS OF PRE- AND POST-ASSESSMENTS

	Pre-Assessment		Post-Assessment	
	<i>Mean</i>	<i>Standard Deviation</i>	<i>Mean</i>	<i>Standard Deviation</i>
Students' Literacy Level	4.05	1.05	5.18	0.96

V. DISCUSSION

The goal of testing was to compare the traditional TaRL program, which uses paper to assess students and relies on instructors to organize and collect student progress data, to the addition of a technological tool for assessments and data collection into TaRL.

A. Pre/Post Assessment Data

In the April 2021 bootcamp, all students in the control and the experimental groups, improved by at least one literacy level over the course of the implementation.

Using the control group and experimental data, a difference of two means t-test was used to determine whether the difference in learning gains between the groups was statistically significant. With a null hypothesis that the learning gains between the groups are not statistically different and an alternative hypothesis that the learning gains between the groups are statistically different, the t-test failed to reject the null hypothesis at a 0.05 significance level. Therefore, there was not a statistically significant difference in learning gains between the control and experimental group at a 0.05 significance level.

The results of comparing the paper and digital assessments were not conclusive, most likely due to poor sampling for the two student groups. The distribution of learning levels in the control group was heavily skewed towards more advanced levels (e.g., paragraph and story), while the distribution of

learning levels in the experimental group was more evenly distributed. In the future, a preliminary assessment determining students' baseline literacy levels will be conducted to assign a representative sample of students more evenly at various literacy levels to the control and experimental groups.

VI. CONCLUSION

TaRL programs can benefit from a technologically aided implementation that optimizes an instructor's process by significantly reducing the amount of time assessing and compiling data and increasing the amount of time where they are promoting learning with students. The Nyansapo platform is intended to aid instructors to that end; although, more work needs to be completed to verify and quantify the efficiency of such digital tools.

A. Summary of Findings

In the April 2021 boot camp conducted with an experimental group and a control group using the digital assessment and paper assessment, respectively, there was no statistically significant difference in the outcome of each group. We have proved that our technological implementation of assessments is just as effective as the traditional, paper process, and therefore a comparable and accurate alternative. With further testing which records more information throughout the boot camp, the technological implementation can be proven to be more efficient than the traditional method and potentially boost learning outcomes for students, due to less time being needed to assess students on paper and input and organize data manually.

B. Future Work

To verify the benefits of the digital implementation, data needs to be collected in the following areas: (1) the accuracy of the custom speech-to-text model that was created for the Nyansapo app specifically for Kenyan primary school students; (2) the accuracy of the app model in placing students in correct levels and identifying their strengths and weaknesses; and (3) the efficiency of the app compared to the traditional paper method in reducing time spent on pre and post-assessments. The digital tool created for the TaRL methodology is hypothesized to make the process more efficient and effective by reducing the time taken for pre- and post-assessments as well as streamlining instructor tasks of collecting, organizing, and analyzing data.

Accuracy of the speech-to-text model is defined as the content marked as incorrect by the model that an instructor also independently verifies as incorrect. The more that the model and an instructor agree, the higher the accuracy.

Accuracy of the in-app model for grouping assignments is defined as the comparison between the grouping assignments produced by the app and the assignments produced by an instructor. By assessing students using both methods during the pre- and post-assessment sections of the program, the differences, if any, can be determined and investigated.

Efficiency of the app is defined as the difference in time between the administration of digital assessments compared to the administration of paper assessments. The nature of this statistic is dependent on the number of mobile devices for use,

the number of instructors available to administer assessments, and the number of students in a boot camp.

To this end, more testing needs to be conducted that focuses on the three pillars - assessments, grouping, and costs. To accomplish this, we have designed a testing protocol to be used in a future boot camp.

C. New Testing Protocol

In a following boot camp, students should be assessed twice during the pre- and post-assessment section of the curriculum. One assessment should be taken digitally on the Nyansapo platform, and one should be traditionally administered by an instructor. During each assessment section, there are two key aspects to investigate: time and accuracy. Regarding time, two key questions are, "What is the difference in time taken by individuals on each type of assessment?" and "What is the difference in total time for the entire group on each type of assessment?" Regarding accuracy, the key question is "How accurate is the speech-to-text model in detecting what words a student mispronounces?"

Therefore, the assessment protocol is as follows.

1. Record the number of mobile devices available for students to use with the latest Nyansapo platform installed, the number of instructors, and the number of students.
2. Begin with the digital assessments and begin recording the time for the entire group.
3. Individually record the time taken by each student to complete an assessment from beginning to end. Each instructor needs to manually record the content that a student mispronounces or that they would normally mark as an error.
4. After all assessments are complete, stop recording the time for the entire group.
5. Then, assess all students using the traditional assessment process on paper while recording time as mentioned in steps 2, 3, and 4.
6. For the post-assessment, alternate the order of type of assessments and repeat all steps.

The content that instructors note down as incorrect for each student will be cross-referenced with digital assessment results to see whether the speech-to-text model is marking content correctly on behalf of students.

After both types of assessments are administered, instructors should compare the grouping derived from the traditional paper assessment to the grouping derived from the digital assessment.

In addition, a major question of the digital implementation is "What is the added cost of physical infrastructure needed to transition to a digital implementation: mobile devices, internet hotspots, and any other needs?" Therefore, any costs incurred stemming from digital infrastructure should be recorded.

Determining quantifiable answers to these questions will offer hard evidence of the value of digital tools like the

Nyansapo platform to organizations and schools implementing TaRL.

D. Impact on Nyansapo's Direction

As a new educational technology start-up, we are still working on gathering sufficient data to prove the value of our platform for instructors, students, and non-governmental organizations. Since the April 2021 boot camp, the Nyansapo platform has grown to encompass a numeracy curriculum adapted from the Uwezo protocol, more detailed individual and group progress pages, and an enhanced Session Planning Tool for instructors to utilize.

With further testing and development, the Nyansapo platform will eventually serve as a robust tool that can encompass any curriculum molded by the TaRL methodology of assessing, grouping, and skill-building with activities.

In addition, the Nyansapo platform represents an opportunity to build technological literacy with students who may not have prior experience. Out of the three boot camps conducted from 2019 to 2021, most of the enrolled students hailed from a mix of under-resourced rural or urban areas. During the August 2020 boot camp, when students were asked if they had access to a mobile phone at home, the overwhelming majority responded that they did not. In an increasingly digital world, this early introduction to technology as a learning tool is vital.

The end goal of the Nyansapo platform is to optimize away the repetitive and strenuous tasks of TaRL - administering assessments, data analysis, and session planning - to increase instructor-student interaction and thereby boost learning outcomes.

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Global Schools Forum - Membership Agreement

This Membership Agreement consists of two parts:

Part A: Outline of Commitment

Part B: Terms and Conditions of Membership

Please read both sections carefully and sign where indicated.

A. Outline of Commitment

About GSF

Global Schools Forum (GSF) works to strengthen global education through supporting non-state organisations in developing countries who serve children from low-income backgrounds. We are a membership organisation united by a common vision of a world in which all children are able to access quality education.

GSF is a catalytic platform for education organisations to improve education quality and scale impact. We do this through networking and professional development opportunities, evidence aggregation and generation, support to entrepreneurs to scale evidenced innovations in partnership with governments, and strategic policy engagement.

Currently, GSF's network of 65 member organizations spans 51 countries across Africa, Asia, and South America, and collectively run or support 30,000 schools that provide quality education to over 5 million children. Our members' commitment to collaboration and positive engagement will contribute globally to a more equitable educational system.

Our Vision

A world where **all children have access to quality education.**

Our Mission

Strengthen the education sector by working with non-state organisations in developing countries who are serving children from middle- and low-income backgrounds.

Our Commitment To Our Members

1 *What we offer*

GSF is dedicated to supporting and representing non-state education providers serving low- and middle-income children, to strengthen their role in the education system globally and maximise the number of children accessing high-quality education.

We do this through:

1. ***A Learning Network***

GSF Connections. Our connections will enable you to learn and form partnerships. In the past 18 months, we have facilitated 140 connections between members, funders, and experts.

GSF Learning Opportunities. Participate in our exclusive and carefully curated learning offers on increasing learning outcomes, improving organisation effectiveness, scaling innovative practices and accessing funding.

GSF Communities of Practice. Learn with and from other practitioners on topics such as approaches to foundational learning and building skills for school to work transition.

GSF Annual Meeting. Take part in this multi-day event which in its last edition brought together 175 educators, funders, policy makers and other ecosystem partners.

2. *A platform for evidence*

GSF databases and evidence products. Become a part of our efforts to compile, make sense of, and share disparate evidence in the sector, including on foundational literacy and numeracy, scaling education innovation, education financing, and government partnerships.

GSF standard-setting for cost-effectiveness and assessment. Benefit from guidance on standardized education impact and cost measurement, and assessment standards.

3. *A forum for policy engagement*

GSF Policy Engagement. Learn about the policy processes of all major funder organisations, and how to forge equitable government partnerships.

4. *Further funding and innovation support opportunities*

GSF Funding. Avail flexible funding to scale your innovative practices through our Impact@Scale Labs.

GSF Funder Connections. Further to the funding and coaching available through the Labs, we will connect you to suitable funders that can facilitate scale-up.

GSF Innovation Support Opportunities. Gain access to new partnerships for innovation. Last year, three of our members won prizes in the “Scalable Innovations for Blended Learning” Solveathon.

We are dedicated to constantly improving our offer to our members. We continuously develop and refine our services based on emerging member needs and feedback.

2 *How we act - our values*

Our community is shaped by our values. At GSF, we commit to upholding these values and creating a culture that is in line with them.

We are transparent and inclusive

- Our membership criteria are clear and simple; cost is not a barrier for entry
- We share information to support the efforts of operators, intermediaries and partnering governments
- Our words and actions are characterised by integrity

We are urgent and outcomes-oriented

- Our actions reflect the urgency and importance of our members’ efforts to put children first.
- We provide a high-quality and relevant member offer.

We operate with kindness and humility

- We understand that people with different experiences and contexts can challenge and strengthen our notions of what is possible
- We seek out feedback and look for best practices to continuously improve the quality of our work. We know there are many different solutions to the challenge we’re trying to solve, and we seek counsel and best practices

Your Commitment to Our Community

1 *Commitment to diversity and inclusion*

GSF recognises that our membership represents an incredible diversity of staff and students in terms of race, ethnicity, religion, gender, socioeconomic status, and sexual orientation. As an organisation, we appreciate the different ideas, perspectives and values this diversity brings, recognising that it makes us stronger in service of our shared mission. We value each and every member and their contributions equally.

We want GSF to be a community in which all our members are welcomed, included, respected, valued, and enabled to fully participate. We expect our members to understand GSF's commitment to diversity and inclusion, and to commit to operating in line with this within the GSF community.

In addition, GSF members are expected to adopt the highest standards of fairness and integrity and no member shall act in a manner detrimental to the interests of students, school communities, the wider public, other members or GSF.

2 *Commitment to participation*

GSF is a community of like-minded members that are interested in discussing, sharing, and collaborating to strengthen schools, school networks, and the non-state sector more broadly.

Our network is at its most dynamic and thriving when its members are engaged and willing to collaborate for the mutual benefit of the sector. GSF therefore expects that its members will:

- Participate and contribute to discussions, events, and forums
- Bring their knowledge and expertise to GSF and actively share best practice with other members
- Disseminate information and learnings within their schools and networks to maximise impact
- Be ambassadors of GSF and invite others to join
- Make recommendations to improve existing services or suggest new ones

B. Terms & Conditions of Membership

Membership is voluntary and GSF is not a governing body for schools or a regulator of member organisations. Membership is organisational not individual. While GSF recognises the autonomy of schools and school networks, members that join GSF undertake to comply with the conditions and requirements set out below.

1. New membership

Membership is open to pre-primary, primary and secondary schools, and school networks in low- and middle-income countries serving children from low-income backgrounds and to intermediary organisations working with schools like these. Requests for membership are reviewed by the GSF team for eligibility based on the information provided in the membership form, gained from the due diligence phone call and references.

If concerns are raised that the behaviour of the applicant is detrimental to their schools (or portfolio of schools) staff, students, the system more widely or any other issue that raises concerns for the Board, then further information may be required during the application process. Please note that membership is at the discretion of GSF leadership.

Once membership has been approved, members have access to the full range of products and services offered by GSF and will be invited to participate in all GSF offers.

2. Membership renewal

GSF membership is automatically renewed annually, until members request to cancel it.

Implementation of a Child Protection Policy within the first year and completion of the GSF Annual Survey/Census are both conditions of GSF membership. If these conditions are not met, membership will not be continued.

3. Membership cancellation

If a member requests to cancel their membership, this takes immediate effect from the date the cancellation request is received in writing by the GSF team, and all member benefits and rights cease at that time.

GSF reserves the right to cancel membership at any time for reasons that contravene this agreement, including but not limited to cases of child abuse or neglect, harassment, fraud, etc. In cases such as these, a review process will be undertaken prior to cancellation and the member will be informed of this.

4. Annual fees

GSF intends to be partially self-sustainable through the annual collection of membership fees. GSF is planning to introduce membership fees in the future. Fees will be tiered based on the self-reported operating budget of the member and will be paid annually. New members receive free membership for the first year, requiring payment in subsequent years. Members will be provided clear guidance on the fee structure at least six months prior to the introduction of the annual fee payment and can decide to continue with GSF at that time.

5. Privacy and Information

All information shared with GSF will be held according to our [Privacy Notice](#) and the sharing guidelines detailed below.

GSF is committed to protecting the personal data of its members. GSF does not provide member specific data to external partners unless prior written permission is granted by the member, except for general information that is already disclosed and available in the public domain (such as organisation logo, name

of CEO/Founder, countries of operation, total schools in network/portfolio, number of students in network/portfolio, hyperlink to member's website).

5.1. Membership form

All members complete a membership application form. Members agree that basic information such as organisation logo, name of CEO/Founder, countries of operation, total schools in network/portfolio, number of students in network/portfolio, and a link to the member's website, can be shared on the GSF website.

5.2. Annual Census

Members are expected to complete the GSF annual census by March/April each year so GSF can collate, analyse, and share the data anonymously on the GSF website and with the sector.

GSF reserves the right to share fully anonymised data collected through the Annual Census. Please see GSF's [Privacy Notice](#) for more information.

5.3. Documentation and policies

5.3.1. Child Protection Policy

GSF has a Child Protection policy, which details its commitment to ensuring the safety and protection of all children. In addition, GSF requires all members to have the right structures and processes in place to ensure the safety and protection of all children directly or indirectly in their care.

In signing this agreement, you will also be asked to confirm that you have an active child protection policy for your organisation.

5.3.2. Admissions Policy

If a member has selective admissions to any of their schools, they will also be required to share their admissions policy as part of the membership application process.

5.3.3. Other

Other documentation may be requested if there are concerns raised about members.

5.4. Information to non-members

Members will ensure that products, information, and documents received from GSF, other GSF members or via any GSF collaborative platforms are not directly or indirectly disclosed or shared with non-member organisations without the written permission of the person or organisation who shared, delivered or provided them.

5.5. Communication

GSF seeks to communicate transparently and regularly with members, enabling them to maximise their membership experience and actively engage in the GSF community. Communication will comply with the latest Data Protection and GDPR requirements and will be for the legitimate purposes of the membership.

5.6. Use of media

On occasion GSF will request the use of photographs or other media from members for our internal and external communication. When using photos from Members, GSF will request written confirmation from the Member that they have appropriate consent for the images they are sharing to be used in marketing and publicity. Any images used shall respect each child's dignity and identity. This means that images will be anonymised in so much as the children will not be named or will be given pseudonyms, and their specific location will not be listed.

GSF will only use images of children where we are confident that appropriate consent has been gained from the organisation sharing the photos with us or where the photos were taken.

6. School registration and licensing

GSF is fundamentally committed to working alongside government. It is the expectation that members comply with relevant national and local school registration systems. Members that are running schools are required to be registered and licensed to operate their schools in their country(ies) of operation. Members that are Intermediary Organisations are expected to know which portfolio schools are registered to operate and be supporting the remainder to register where possible. Membership can be granted in cases where compliance with licensing requirements and positive engagement with government are demonstrated even if licenses are not yet in place. If concerns are raised about members' compliance with government regulations, GSF will use discretion to inform their decision-making processes.

7. Advertising

Members will ensure that advertising, articles, speeches, notices, or any forms of publication are governed by professional ethics.

8. Governance and management

The GSF Board retains the leadership and decision-making responsibility. GSF is managed by a small team, headed by the CEO and reporting to the Board; the GSF team will submit Progress Reports to the GSF Board, at quarterly Board meetings.

- 8.1. The Board currently comprises representatives from Ark, Kaizenvest, IDP Foundation, Teach For All, UBS Foundation, World Bank as well as from three GSF member organisations. Members will be advised of governance changes and may be invited to join the Board as member representatives. These terms and conditions will be updated at the time of change.

9. Liability

GSF does not take responsibility for the actions of members or any of its affiliates, directors, officers, employees or agents (the "**Operators**"). GSF shall not be liable for any claim, demand, action, liability, damages, cost, loss or expense arising from any actions of the Operators (including, without limitation, legal fees, costs and expenses). Membership of GSF is not necessarily an endorsement of an organisation.

GSF has zero tolerance for corruption, discrimination, and mistreatment of staff and students. If GSF is notified of such activities, or other behaviours and situations that raise concerns, in member organisations the GSF Board would take immediate recourse. If any behaviour or action of a GSF member brings GSF into disrepute, the GSF Board will take the necessary steps to protect the reputation and work of GSF.

10. GSF representation

GSF is not a trade association and does not represent individual members. Member organisations cannot refer to GSF in the event of a conflict with other members, and the expectation is that the GSF name would not be used beyond what can be reasonably expected.

11. Updates to this document

GSF will regularly review this document and update it as the organisation grows and develops.

CHILD PROTECTION POLICY CONFIRMATION

Please read and initial the statement below to confirm that it applies to your organisation:

Confirmation	Initials
We (the member) have Child Protection policies and procedures in place that are appropriate for the context in which our organisation operates.	MM

CONFIRMATION OF MEMBERSHIP

In signing this agreement, we confirm that we wish to become members of GSF. In line with GSF's vision and mission, we will play our part in ensuring all children, regardless of their background, are able to access quality education.

We have read and understand the **Outline of Commitment** and agree to uphold, within the community, the expectations of members outlined there:

- We understand GSF's commitment to diversity and inclusion, and we will operate in line with this within the GSF community.
- We commit to participating in the GSF community; contributing, sharing and learning together with other GSF members to strengthen the non-state education sector through our membership of GSF.

We have read and agree to the **Terms & Conditions** stated above and confirm that we will disclose to GSF any major Child Protection issues that occur during our period of membership.

Name of Organisation applying for membership: **The Nyansapo Foundation**

Name(s) of Individual Appointees (Typically members of the Senior Management Team):

1 **Mumbe Mwangangi**

2 **Edward Amoah**

3



NYANSAPO AI



ANNUAL REPORT

2022/2023

info@nyansapoi.net
www.nyansapoi.net



Prepared by
Edward Idun Amoah

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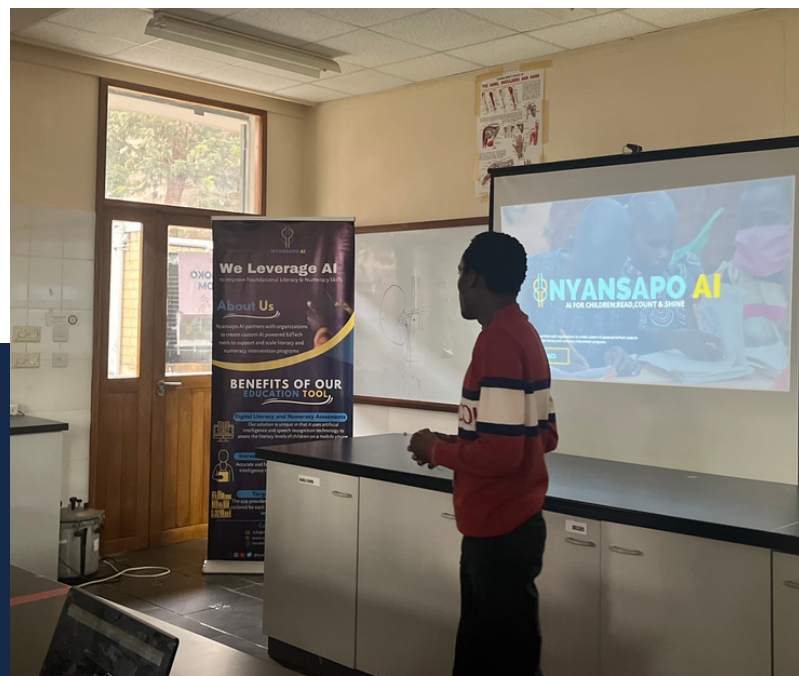


Profile Of Company

Nyansapo AI is a social enterprise with the vision of globally transforming foundational literacy and numeracy (FLN) education with an AI-powered platform. The Nyansapo AI platform includes an AI-powered Nyansapo teaching mobile application and a school management web application.

The year 2023 was our year of development. We joined numerous accelerator and training programs to refine our technology and business model. We joined AWS Build Accelerator and Microsoft for Startup to learn from industry experts and prepare our product for our first public launch. We participated in the Futuris Institute AI4Good training, where we refined our business model and presented it to win the AI4Good Category at the African Startup Conference.

The year 2024 is our year of growth. We seek collaborations and funding to run pilot projects and expand across multiple countries. We begin this year with our official product launch on January 22, which allows everyone to access our platform globally. We are also commencing a pilot project in 20+ primary schools across Bungoma, Kenya, collaborating with collaboration with the Ministry of Education Bungoma County Government and the Amani Community-Based Organization (CBO) with a \$29,000 grant from USAID YALI. We aim to raise \$10,000,000 to expand and run pilots in 100+ primary schools in rural and urban areas in Kenya, Uganda, Tanzania, Ghana, and Nigeria, targeting 100,000 children.



Our Vision

To create a world where every child, regardless of their socioeconomic background, has equal access to quality education and personalized learning opportunities powered by innovative and accessible AI technology.



Our Mission

- 1** To create an affordable and accessible AI-powered teaching and learning platforms for foundational literacy and numeracy education.
- 2** To empower educators with tools to transform education and deliver quality education for all.

OUR TEAM



Mumbe Mwangangi
Co-founder and CEO



Edward Amoah
Co-founder, CTO, and Lead
AI Engineer



Justice Eli
Lead Backend and Android
Engineer



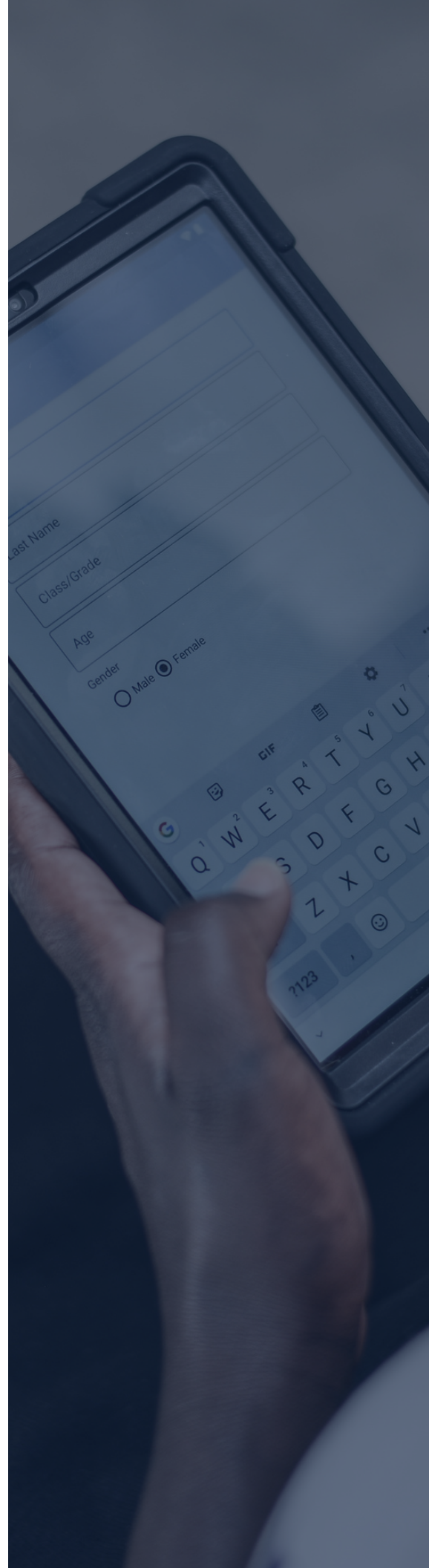
Alex Magwe
Lead Web Engineer and
UX/UI Designer



Albright Mbia
Communications and
Marketing Lead



Faith Atieno
Learning and Partnership
Lead



Advisory Board

Dr. John Gershenson

Director of Humanitarian Engineering and Social Enterprise (HESE) Program, Penn State, Pa, USA.



Dorcas Owinoh

Executive Director of LakeHub, Kisumu, Kenya.



Dr. Brad Zdenek

Director of Economic Development and Student Programs, Penn State, Pa, USA.



HIGHLIGHT IN 2023



DIGITAL INNOVATION GALA, MARCH 20TH

Qualified to showcase our platform at the digital innovation gala organized by Save the Children Kenya & Madagascar.



EDTECH EAST AFRICA, SEPTEMBER 20TH

We were recognized and got the opportunity to present our platform at the EdTech East Africa Summit 2023.



USAID YALI GRANT, OCTOBER 11TH

\$29,000 grant to run pilot programs across 20+ primary schools in Bungoma, Kenya, in collaboration with the Amani CBO, our implementation partner.



ACCELERATOR PROGRAMS, NOVEMBER 28TH

Accepted into the AWS Build accelerator and Microsoft for Startups, where we have access to \$100,000 AWS cloud credits and \$150,000 Azure credits, respectively for development and scaling.

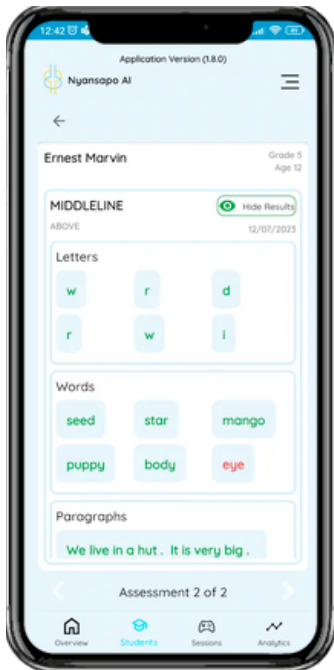


AI4GOOD COMPETITION, DECEMBER 6TH

We were privileged to participate in the Futuris Institute AI4Good training and competition. We were awarded 1st place as the most promising AI startup in Africa at the African Startup Conference in Algiers, Algeria.

OUR PLATFORM

The Nyansapo AI platform consists of an AI-powered Android mobile application for teaching and a web application for school and classroom management.

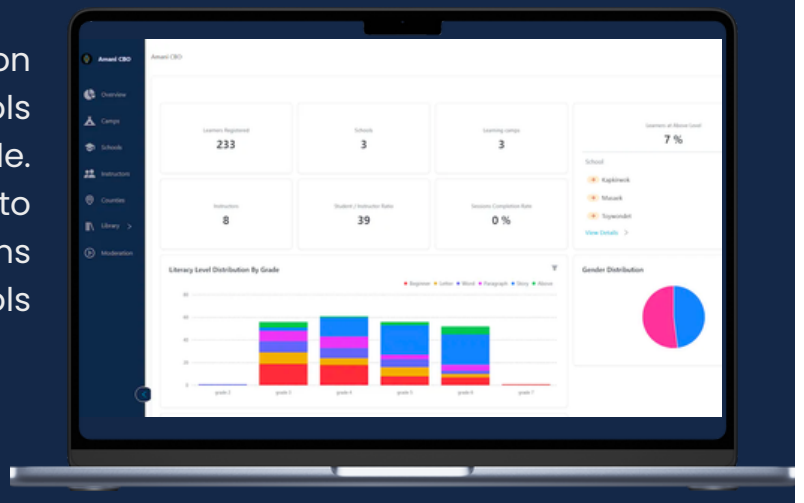


NYANSAPO TEACHING

The mobile application empowers teachers to implement foundational literacy and numeracy (FLN) programs in their classrooms using the teaching at the right level (TaRL) methodology. The mobile application utilized AI to automate formative assessment to identify learning gaps, saving the teacher time and hustle. The mobile application also utilizes a custom generative AI to create tailored session plans for each classroom according to the children's learning gaps. This saves the teacher time and, more importantly, ensures that every child in the classroom gets the teaching they need.

NYANSAPO MANAGEMENT PLATFORM

The management web application empowers organizations and schools to implement FLN programs at scale. The web application enables people to manage and monitor all FLN programs implemented across multiple schools and classrooms.



WHATS AHEAD



2024 is our year of growth. We aim to grow and expand the impact of our organization to reach more children across Africa. We aim to raise over \$10,000,000 capital to support our project as we expand into at least five African countries. We aim to run pilot programs in over 100+ primary schools in rural and urban areas to continue iterating and create a sustainable and effective tool for teaching and learning. We aim to reach and improve the foundational literacy and numeracy of over 100,000 children by the end of the year.

Finally, we aim to continue learning from the market and establish a sustainable business model to keep this business self-sufficient and sustainable. By the end of the year, we aim to have 100+ primary schools enrolled as paying customers on our platform.



\$10,000,000 Capital

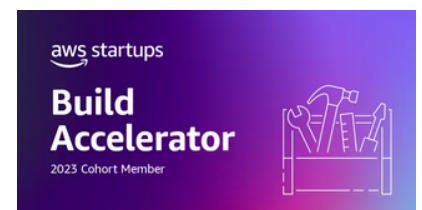
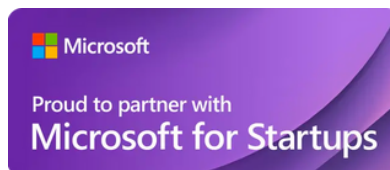


Scale to 100+ schools



Sustainable Business

Global Partners



Call for Action

In our year of growth, we seek collaborations and funding. Any NGO, CBO, or organization worldwide that aims to improve children's foundational literacy and numeracy can adopt our platform and partner with us to run foundational literacy and numeracy programs in their community. Go to our website today to get started with us. Contact us at info@nyansapodai.net for support and collaboration.

Anyone passionate about improving education for children can support or donate to our vision directly on our website by the end of January. Nyansapo Labs Inc. is registered as a 501(c)(3) organization and accepts charitable donations.



Collaborate with us



Donate for our vision

NYANSAPO AI

Organizational Chart



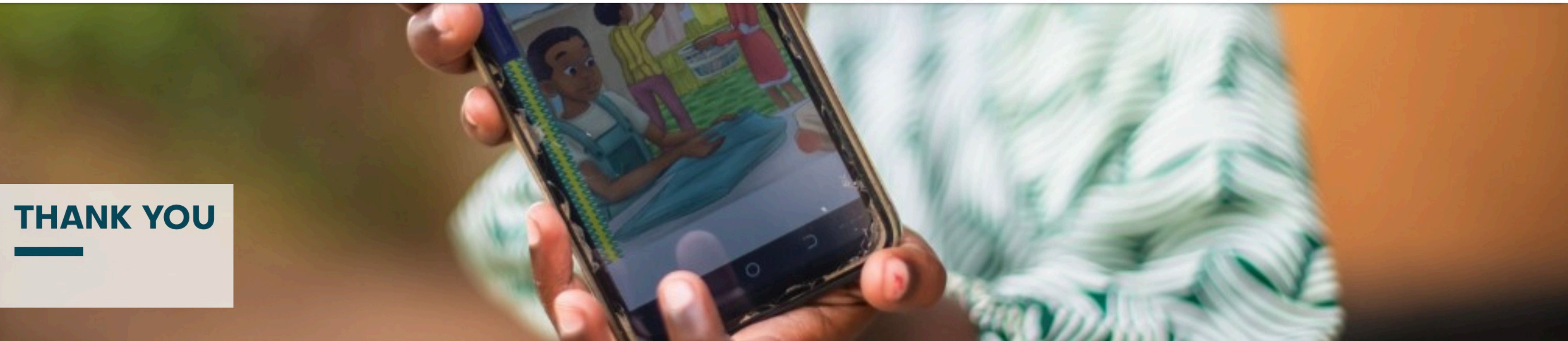
Advisory Board





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NABU

Thank you for taking the time to complete the membership application form. Your submission has been successfully received. We appreciate your interest in joining the community, we shall carefully review your application and get in touch with you for the next steps.

Should you have any questions, or need further assistance, please feel free to contact [Mercy Karumba](#).

We are a dynamic community working to improve education outcomes at scale.

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[CHILD PROTECTION POLICY](#)
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Child Protection Policy

Effective date: 6/11/2023- 12/12/2024

1. Introduction

Nyasapo AI ("the Company") is committed to safeguarding the well-being and protecting the rights of children who interact with our services. This Child Protection Policy outlines the guidelines and procedures we follow to ensure the safety and security of children in accordance with international child protection laws, as well as applicable Kenyan data protection laws and regulations.

2. Definitions

- a. **Child:** Refers to any person under the age of 18 years.
- b. **Child Abuse:** Includes any form of physical, emotional, sexual abuse, neglect, or exploitation of a child.
- c. **Child Protection:** Refers to the measures taken to prevent and respond to child abuse, promote child well-being, and ensure the rights of children are respected and upheld.

3. Compliance with International and Kenyan Laws

- a. The Company adheres to international child protection standards, including but not limited to the United Nations Convention on the Rights of the Child (UNCRC) and the International Labor Organization (ILO) conventions related to child protection.
- b. In addition, the Company complies with relevant Kenyan laws and regulations concerning child protection and data protection, including the Children's Act and the Data Protection Act.

4. Responsibilities

- a. The Company's management, staff, contractors, and partners have a collective responsibility to prioritize child protection in all aspects of our operations.
- b. Appropriate training and awareness programs are conducted to ensure all individuals associated with the Company understand their responsibilities and obligations regarding child protection.

5. Risk Assessment and Mitigation

- a. The Company conducts regular risk assessments to identify potential risks and vulnerabilities related to child protection within our operations.



b. Measures are implemented to mitigate identified risks and vulnerabilities, ensuring the safety and well-being of children.

6. Consent and Confidentiality

a. When collecting and processing personal information of children, the Company obtains appropriate consent from the child's parent or legal guardian in accordance with applicable Kenyan data protection laws.

b. Confidentiality of children's personal information is maintained, and access to such information is restricted to authorized personnel only.

7. Reporting and Response

a. Any suspicions, allegations, or incidents of child abuse or inappropriate conduct involving a child within the Company's operations must be reported immediately.

b. The Company has designated child protection focal points to handle reports, conduct investigations, and take appropriate actions in accordance with established protocols.

c. Reporting mechanisms are accessible and known to all employees, contractors, partners, and stakeholders.

8. Training and Awareness

a. The Company provides ongoing training and awareness programs on child protection to all relevant personnel, emphasizing their roles and responsibilities in safeguarding children.

b. Educational materials and resources are made available to promote awareness and understanding of child protection principles.

9. Cooperation and Collaboration

a. The Company actively collaborates with relevant authorities, organizations, and stakeholders to strengthen child protection measures and respond effectively to child protection concerns.

b. Cooperation with law enforcement agencies is ensured, as required, to address child protection issues.

10. Review and Update

a. This Child Protection Policy is reviewed periodically to ensure its continued relevance and compliance with evolving international and Kenyan laws and regulations.

b. Updates and revisions to the policy are communicated to all relevant personnel within the Company



11. Compliance and Accountability

a. Non-compliance with this Child Protection Policy may result in disciplinary action, including termination of employment or contractual agreements, in accordance with applicable laws and regulations.

If you have any questions or concerns regarding this Child Protection Policy, please contact mumbe@nyansapoi.net.

By implementing this Child Protection Policy, Nyasapo AI reaffirms its commitment to providing a safe and secure environment for children and upholding their rights as outlined in international and Kenyan laws and regulations.



DECLARATION

**I..... of NATIONAL ID/PASSPORT
NO.....PHONE NUMBER.....EMAIL ADDRES
S.....**

**HAVE READ AND UNDERSTOOD AND AGREE THAT WHILE IN
A RELATIONSHIP WITH NYANSAPO AI AND WHILE IMPLEMENTING
WITH THE NYANSAPO AI'S ACTIVITIES AND WHILE IN CONTACT
WITH CHILDREN WHO ARE INVOLVED IN NYANSAPO AU ACTIVITIES,
WILL ABIDE TO THIS POLICY AND THAT IN VIOLATION TO THIS,
BY COMMISSION OR OMISSION NYANSAPO AI TOGETHER WITH
OTHER LEGAL ENTITIES SHALL INSTITUTE A LEGAL PROCESS.**